



HEALTH SYSTEM PHARMACISTS

Evaluation of Direct Oral Anticoagulant Use and Implementation of Pharmacist-led Direct Oral Anticoagulant Telephone Clinic in an Outpatient Federally Qualified Health Center

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Objectives

- Review recommendations for Direct Oral Anticoagulation therapy based on CHEST guidelines.
- Discuss process for implementation of pharmacist-led telephone clinic.
- Assess the prescribing patterns around direct oral anticoagulants (DOACs) in a primary care setting.

Statement of disclosure

- I have no conflicts of interest to disclose.

Background

- Thromboembolic events such as cardioembolic stroke and venous thromboembolism (VTE), which includes deep vein thrombosis (DVT) and pulmonary embolism (PE), are major causes of morbidity and mortality.¹⁻²
- Historically, warfarin has been the mainstay of anticoagulation therapy for VTE prevention and atrial fibrillation.³⁻⁴

Background

- In clinical trials, DOACs were shown to be non-inferior to warfarin and are suitable alternatives.⁵⁻⁷
- The CHEST guidelines now recommend DOACs over warfarin for oral anticoagulation therapy.⁸⁻⁹

CHEST Recommendations

- “For VTE and no cancer, as long-term anticoagulant therapy, we suggest **dabigatran (Grade 2B), rivaroxaban (Grade 2B), apixaban (Grade 2B), or edoxaban (Grade 2B)** over vitamin K antagonist (VKA) therapy, and suggest VKA therapy over low-molecular-weight heparin (LMWH; Grade 2C)”.⁸
- “In patients with AF who are eligible for OAC, we recommend **NOACs** over VKA(Strong recommendation, moderate quality evidence)”.⁹

Background

- Although DOACs have provided possible solutions to several challenges associated with warfarin therapy, some prescribers remain unaware of
 - Potential drug-drug interactions
 - Dose adjustments based on renal function, age, weight, indication
 - Adherence issues
 - Cost implications

Purpose of the Study

- To evaluate the prescribing patterns around DOACs in an outpatient teaching clinic and to implement a pharmacist-led intervention to potentially improve patient outcomes.

Adapted Medication Appropriateness Index (MAI)¹⁰⁻¹¹

Criteria	Category	Instructions
Indication	A	Valid indication exists
	B	DOAC used as a last resort or indication does not fit within reimbursement criteria
	C	Off-label use
Choice	A	DOAC preferred choice: labile INR with VKA, CI to VKA, patient preference, recurrent stroke/VTE on VKA, resistance to VKA
	B	No contraindication, not yet tested, no recurrent stroke/VTE on VKA
	C	Not preferred choice: severe renal insufficiency, poor compliance, need for drug monitoring, severe hepatic impairment, recurrent VTE/stroke on current DOAC
Dosage	A	Receives daily dose as recommended
	C	Inappropriate daily dose (too low or too high)
Correct administration	A	Correct modalities of DOAC intake
	B	Limited clinical relevance for modalities of DOAC intake (timing of DOAC administration, DOAC given with or without food)
	C	Inappropriate modalities of DOAC intake (daily vs. twice daily)
Practical administration	A	No difficulties taking dosage form
	C	Difficulties taking the drug (BID dosing, not able to swallow capsules)
Drug-drug interactions	A	No DDI
	B	Potential DDI (caution or warning) without s/sx adverse event
	C	Contraindicated with disease, presents high risk, or DOAC used with caution and positive s/sx of worsening disease
Drug-disease interactions	A	No drug-disease interactions
	B	Potential interaction (caution or warning) without s/sx of adverse event
	C	Contraindicated with disease, presents high risk, or DOAC used with caution and positive s/sx of worsening disease
Therapeutic duplication	A	DOAC is the only antithrombotic
	B	Concomitant anticoagulation when switching therapy (DOAC to VKA)
	C	Duplication of antithrombotic
Duration of therapy	A	Duration in accordance with manufacturer indication
	C	Duration not appropriate based on recommendation

Methods

DOAC Utilization

- The electronic medical record was used to identify patients who have received dabigatran, rivaroxaban, apixaban, or edoxaban as an active medication at PCHC.
- Inpatient and outpatient records were reviewed to identify any adverse reactions such as hemorrhagic and/or thrombotic events, hospital admissions, or emergency room visits related to anticoagulation use.
- An adapted medication appropriateness index (MAI) was utilized to evaluate inappropriate prescribing.

Pharmacist Intervention

- A pharmacist-led telephone clinic will then be implemented for patients who are actively prescribed a DOAC and who provide consent.
- The organization's anticoagulation questionnaire that is used to assess patients taking warfarin will be administered over the phone to assess adherence, tolerability, and possible adverse effects. Counseling will also be given regarding appropriate use and continued therapy.

Patient follow-up

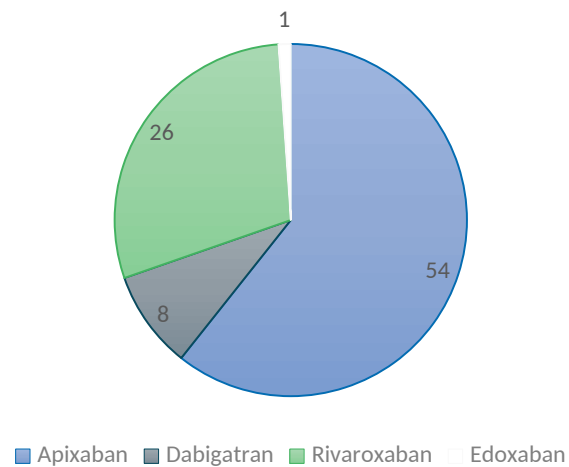
- Primary endpoint: hemorrhagic and/or thrombotic events
- Secondary endpoints: Appropriateness of therapy based on MAI
- The primary and secondary endpoints will then be evaluated and compared to the standard of care without pharmacist intervention.

Baseline Characteristics

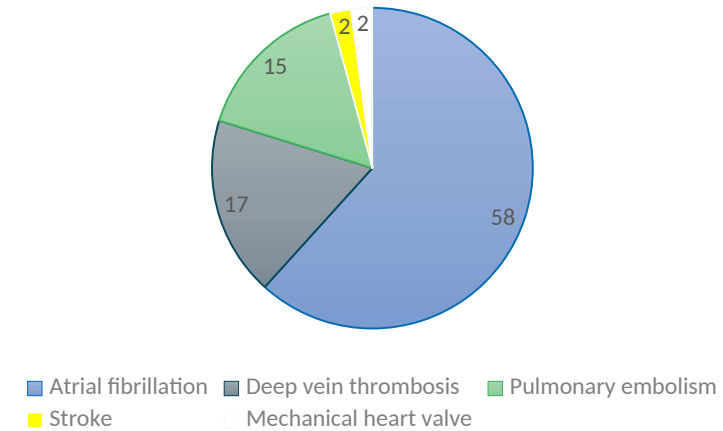
Baseline Demographics	
Characteristic	PCHCenter (n=89)
Age (years; average)	68
Sex	
• Male (%)	44.9
• Female (%)	55.1
Race	
• Caucasian (%)	97.8
• African American (%)	1.1
• "Other" (%)	1.1
CHA ₂ DS ₂ -VASc (atrial fibrillation; average)	3.75
HAS-BLED score (atrial fibrillation; average)	1.3

Baseline Characteristics

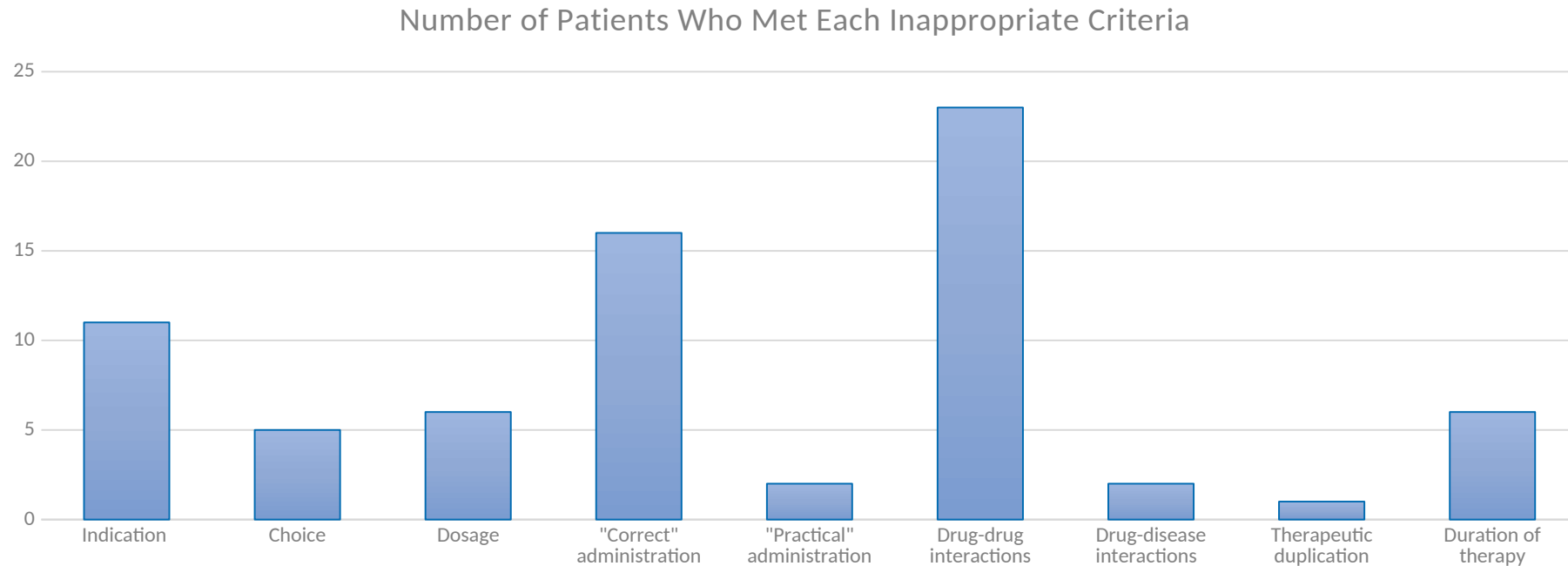
Number of Patients Taking Each DOAC



Number of Patients With Each Indication



Review of Inappropriate Criteria



Retrospective Review without Pharmacist Intervention

- Primary endpoints:
 - Hemorrhagic Events: 10.1%
 - Thrombotic Events: 6.7%
- Secondary Endpoints:
 - 57.3% patients met at least 1 inappropriate criteria

Preliminary Results

- Primary Endpoints:
 - Hemorrhagic events: 12.5%
 - Thrombotic events: 6.3%
- Secondary Endpoints:
 - 37.5% patients met at least 1 inappropriate criteria.

Next Steps

- Continue enrolling patients in the pharmacist-led telephone clinic
- Follow-up with patients once a month to:
 - Assess appropriateness in therapy
 - Monitor adherence
 - Provide patient education
- Collect data then compare to standard of care before pharmacist intervention

Assessment Question

- What was the most common indication for DOAC use among patients in this study?
 - a) Deep Vein Thrombosis
 - b) Pulmonary Embolism
 - c) Atrial Fibrillation
 - d) Mechanical Heart Valve

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References

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Questions?

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